William Cook

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Education

Liberty University Expected May 2028
B.S. Computer Science (Software Engineering & Technology Entrepreneurship) *Lynchburg, VA*
GPA: 3.78 / 4.0 · **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Architecture, Database Design, Information Security, Network Architecture, Advanced C++

Experience

Cole Design & Development LLC Dec. 2025 – Present
Software Engineering Intern *Hudson, OH*
Node.js, Express, React, PostgreSQL, MQTT, Docker, AWS EC2, Python, Rust/Tauri

- Built ColeCloud, an in-house alternative to a paid IoT platform, as the only developer. Had it live in three weeks; it now runs in production with live dashboards, device maps, telemetry history, reports, and alerting.
- Designed the backend around JSON config and PostgreSQL JSONB, so all four device types share the same code and adding a new one is mostly a config entry.
- Gave every device its own X.509 certificate over mutual TLS, then audited the full stack and fixed what it turned up: an anonymous MQTT WebSocket open to the internet, API routes with no login, and runaway logs that took down Postgres.
- Wrote Rust/Tauri commands that let our desktop app configure hardware directly, and cleaned up the CRC and message-parsing code in a Python device simulator.

Missionize June 2025 – Sept. 2025
Full Stack Developer *Remote*
React Native, Python, FastAPI, MongoDB

- Built the habit tracking, journaling, and dashboard screens in React Native for Discipler.io, an AI app for church ministries, along with the REST endpoints behind them.
- Worked on a six-person startup team using feature branches, pull requests, and code review.

Projects

Self-Hosted Home Server, *personal project* Ongoing
Linux, Python, Nginx, systemd, Tailscale

- Run five services on a Raspberry Pi 5 with 1TB of NVMe storage: three Minecraft servers, a Discord bot, and a custom Magic: The Gathering app.
- Set up Nginx with HTTPS, key-only SSH, Tailscale for remote access, systemd for the services, and automated backups.
- Wrote async Python services that process logs and forward events.

Leadership at Liberty University

Japanese Language Club, *President* Aug. 2026 – Present

- Lead a club of 25+ members and plan our meetings and events around Japanese language and culture.
- Built the club website, lujapanese.com, and host it on a self-managed Linux server.

Competitive Programming Club, *Officer* Aug. 2024 – Present

- Teach weekly C++ and Python sessions for 20+ members on data structures, algorithms, and contest strategy.
- Solved 150+ problems on Kattis while preparing for ICPC.

Awards

ICPC Mid-Atlantic Regional, Division 1 – tied for 20th of 111 teams 2025
ICPC Mid-Atlantic Regional, Division 2 – 6th of 61 teams 2024

Skills

Languages: C++, C, Python, JavaScript, TypeScript, Rust, SQL
Frameworks: React, React Native, Node.js, Express, FastAPI, Tauri
Data & Infrastructure: PostgreSQL, MongoDB, MQTT, Docker, AWS (EC2, S3), Linux, Nginx, Git